

Software Requirements Specification (SRS) Document for DemoQA Website

1. Introduction

1.1 Purpose

This document delineates the Software Requirements Specification (SRS) for the DemoQA website, serving as a definitive reference for developers and testers. The primary objective is to ensure that the website's design, development, and testing align with the stated requirements and functionalities.

1.2 Scope

DemoQA provides a sophisticated web-based platform showcasing interactive UI elements, enabling users to engage with functionalities such as forms, widgets, alerts, and interactions. This platform is a valuable resource for developers and testers seeking to validate automation scenarios.

Core Features:

- An intuitive, interactive interface.
- Comprehensive support for automation testing validation.
- Categorized functionalities: Elements, Forms, Alerts, Widgets, Interactions, and Book Store.

1.3 Audience

This document addresses the following stakeholders:

- **Developers:** To comprehend and implement functional and non-functional requirements.
- **Testers:** To facilitate the creation of exhaustive test cases.
- **Project Managers:** To oversee requirement adherence throughout the development lifecycle.

1.4 References

- DemoQA Website: <https://demoqa.com/>
- Selenium and Robot Framework Documentation

2. System Overview

The DemoQA system is a web-based application comprising multiple interactive modules. It is designed to support the learning and execution of web automation testing practices.

3. Functional Requirements

3.1 Elements Page

- **Features:**
 - Text Box: Accepts and submits user input.

- Check Box: Facilitates multi-item selection.
- Radio Button: Enables single-item selection.
- Web Tables: Permits add, edit, delete, and search operations.
- Buttons: Interactive button functionality testing.
- Links: Includes functional and broken links.
- Upload and Download: Supports file interactions.
- Dynamic Properties: Elements with dynamically changing attributes.
- **User Interaction:**
 - Testing includes text input, clicking actions, file operations, and dynamic behavior.

3.2 Forms Page

- **Features:**
 - Practice Form: Validates user data inputs across multiple fields.
- **User Interaction:**
 - Validation and submission of form entries.

3.3 Alerts, Frames, and Windows Page

- **Features:**
 - Alerts: Includes confirmation, timed, and simple alerts.
 - Frames: Nested and side-by-side frame management.
 - Modal Dialogs: Tests modal-based popup behavior.
- **User Interaction:**
 - Scenarios include alert handling, frame switching, and modal interaction.

3.4 Widgets Page

- **Features:**
 - Includes Accordion, AutoComplete, Date Picker, Progress Bar, Tabs, ToolTips, and Menu.
- **User Interaction:**
 - Validates element-specific behaviors like date selection and menu navigation.

3.5 Interactions Page

- **Features:**
 - Drag and Drop: Tests movement of elements to target areas.
 - Selectable, Resizable: Focuses on element resizing and selection behaviors.

- **User Interaction:**
 - Covers drag-and-drop functionality and element resizing.

3.6 Book Store Application Page

- **Features:**
 - Search by book title or author.
 - Manage user collections and account credentials.
 - **User Interaction:**
 - Tests book search functionalities and account management.
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4. Non-Functional Requirements

4.1 Performance

- Pages should render within 2 seconds under typical conditions.
- The platform must maintain responsiveness across devices.

4.2 Usability

- The interface should accommodate novice users effectively.
- Interactive elements must provide intuitive feedback (e.g., form validation errors).

4.3 Compatibility

- Fully supported on Chrome, Firefox, Safari, and Edge.
- Ensures responsiveness for mobile and desktop devices.

4.4 Security

- Enforces secure data handling and input validation.
 - Implements rigorous sanitization protocols for user inputs.
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5. Requirement Elicitation for Testers

5.1 Inputs for Test Design

- **Interactive Elements:** Comprehensive testing of UI components, including forms and modals.
- **User Scenarios:** Specific focus on scenarios like form submission, drag-and-drop, and alert handling.
- **Error States:** Validation of error handling for broken links and invalid inputs.
- **Dynamic Behavior:** Verification of elements such as progress bars and dynamically changing properties.

5.2 Traceability Matrix

Mapping requirements to test cases ensures complete coverage:

Requirement ID Feature		Test Case ID
FR-01	Text Box Submission	TC-01-001
FR-02	Handling Alerts	TC-03-005
FR-03	Drag and Drop Interaction	TC-05-003
FR-04	Book Store Search Functionality	TC-06-002

6. Conclusion

This SRS document articulates the requirements for the DemoQA website comprehensively. By outlining both functional and non-functional specifications, it ensures robust development and validation processes. The structured framework supports complete test coverage and efficient test case design.